Multidimensional Remote Sensing Change Detection Based on Siamese Dual-Branch Networks

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***Abstract*— Deep learning models, particularly convolutional neural networks (CNNs), have demonstrated outstanding feature learning capabilities, leading to remarkable performance in remote sensing change detection (RSCD) tasks. However, their most critical drawback lies in the lack of effective modeling of global information. This deficiency affects the model's understanding of the overall context and structure of the entire image, making it difficult to distinguish between background and target areas, thereby leading to the erroneous identification of change regions. Second, features extracted by traditional backbone networks contain a significant amount of noise, resulting in blurred boundaries of changed objects. The challenge of effectively fusing detailed and semantic information to accurately differentiate pseudo changes remains significant. Furthermore, how to fully exploit multiscale information is another issue worth considering. We propose a full-scale multidimensional interaction network called SDSN, which enhances feature representation by leveraging both detail and semantic branches. Initially, bi-temporal images are processed by the encoder to extract coarse multiscale features. The semantic branch guides shallow-scale features, while the detail branch focuses on deep-scale features. Multikernel receptive module (MRM) aggregates global information. The detail branch utilizes a diversity variance module (DVM) and differential operations to generate refined change maps with noise reduction and background suppression. A multidimensional cross-perception module (MCM) guides the fusion of these change maps, establishing multidimensional dependencies to enrich feature representation. Compared with previous methods, SDSN demonstrates greater performance under complex environmental conditions, particularly noteworthy for its fewer parameters (4.03 M) and lower computational costs (7.94 G). The code is publicly available at https://github.com/dpt000121/dpt.**

***Index Terms*— Change detection (CD), detail branch, image difference, multidimensional cross-perception, remote sensing (RS), semantic branch.**

I. Introduction

AN ESSENTIAL objective of remote sensing change detection (RSCD) entails discerning alterations within remote sensing (RS) imagery captured at two or more distinct time instances, including where the changes have occurred, the types of changes (e.g., increase in buildings, decrease in forests, and so on), and the extent of the changes. This kind of information is important for land management [1], urban and rural development surveys [2], [3], disaster assessment [4], [5], and other fields. Consequently, a plethora of techniques have emerged, broadly categorized into two groups: conventional approaches and those rooted in deep learning methodologies. The mainstream approach of early traditional methods involved using manually crafted features to capture variations. Nowadays, deep learning has garnered widespread attention for its outstanding performance in computer vision. For the task of RSCD, many deep learning methods have emerged and are gradually replacing traditional approaches. For example, ICIF-Net [6] integrates the strengths of CNNs and transformers, proposing a dual-branch structure to couple local and global features. It introduces mask-based aggregation and spatial alignment to fuse the capabilities of local feature extraction and global modeling. However, this leads to a significant increase in parameters and computational costs, making it impractical for deployment on small devices such as satellites. To boost the interaction between two images, SARAS-Net [7] conducts prefeature and postfeature differential operations on two input images to pinpoint the altered regions. While the powerful transformer furnishes the model with abundant semantic information, it tends to overlook the capture of local details. CSViG [8] utilizes both horizontal and vertical feature fusion modules, allowing it to capture a broader range of short- and long-distance dependencies, however, it does not consider channel and spatial interaction mechanisms.

The primary goal of this study is to develop an RSCD model that can effectively distinguish between real and pseudo changes by capturing both comprehensive spatial and semantic information. The model aims to maintain high detection accuracy while minimizing the number of parameters and computational overhead. While the lightweight backbone MobileNetV2 [9] has gained attention for its excellent performance, it primarily focuses on local feature extraction, resulting in a limited receptive field. To address this limitation and capture long-range dependencies, the model introduces a semantic branch that focuses on global context information. This branch works in conjunction with a refined detail branch, and their features are fused through cross-dimensional interactions in a multidimensional cross-perception module (MCM), ensuring more accurate and robust change detection (CD). Through the synergy of diversity variance module (DVM), multikernel receptive module (MRM), and MCM, SDSN achieves competitive detection accuracy, e.g., IoU scores of

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72.08%, 89.83% and 84.41% on SYSU [10], BCDD [11], and LEVIR [12] datasets, respectively. The key contributions of this study can be summarized as follows.

1) To address the issue in RSCD where CNNs lack global semantic information, making it difficult to distinguish between background and buildings, we propose the innovative SDSN model. This introduces a semantic branch specifically designed to capture global contextual information, which compensates for the lack of long-range information in the detailed branch, thereby guiding the generation of a change map with both rich semantic and detailed information.
2) In this network, three modules have been proposed. DVM is applied in detail branch to suppress noise and background, refining the blurred boundary issues commonly associated with traditional backbone networks. MRM is applied in the semantic branch to extend the receptive field, providing a more comprehensive understanding of the context with generated semantic change maps. To fully leverage multiscale information, both branches undergo feature enhancement and adjustment before feeding into MCM. MCM effectively integrates semantic and detail change maps, enabling the network to capture comprehensive feature information from various perspectives and across dimensions.
3) We validate our method's superiority through extensive experiments, outperforming thirteen prior methods on SYSU, BCDD, and LEVIR. And achieved fewer parameters (4.03 M) and lower computational costs (7.94 G). The trained model has been uploaded at https://github.com/dpt000121/dpt for evaluation.

The article proceeds as follows. Section II briefly reviews relevant works. Section III outlines our framework. Section IV presents a comprehensive experimental analysis. Section V summarizes the article.

II. Related Work

A. Traditional RSCD Methods

In the past, traditional CD methods relied on manual feature extraction constituted one of the primary approaches. For example, there was algebra-based image differencing [13], image rationing [14], and image quantification [15], which processed RS images through mathematical operations and statistical analysis with the aim of extracting quantitative information in images, such as image metrics, ratios, or eigenvalues for further analysis and decision making. Classification-based support vector machines [16], random forest regression [17], and K-neighborhood (K-NN) [18], models classified each pixel point or each image block to determine whether a change has occurred. Methods based on image transformations mainly use principal component analysis (PCA) [19], tasseled cap transformation (KT) [20], or other methods in order to transform RS images at two-time points so that they were somehow aligned and then detect the differences between the transformed images. However, these typical methods relied heavily on priority domain knowledge and were subject to environmental factors such as season, light, etc., resulting in poor robustness. To address these problems, Xue et al. [21] have proposed a method based on image reconstruction loss (CDRL) using CycleGAN [22] photometric transformation by style-shifting, effectively alleviating the problems caused by environmental changes, but it brought problems such as high computation costs and low robustness. It was not flexible and efficient enough to deal with complex and diverse RS image data, limiting the performance and applicability of RSCD.

B. CNN-Based RSCD Methods

With the development of automatic feature learning methods such as deep learning, modern CD methods prefer to use automatic feature learning techniques such as deep learning, which are able to automatically learn useful features in images without relying on domain experts to manually design features, that means models are capable of capturing more abstract and advanced features, improving the accuracy of detection. Convolutional neural networks (CNNs) [23], [24], [25], recurrent neural networks (RNNs) [26], [27], fully convolutional networks (FCNs) [28], convolutional RNNs (cRNNs) [29], graph convolutional networks (GCNs) [30], [31], generative adversarial networks (GANs) [32], and other deep learning models bring good performance for RSCD tasks. For example, CNNs classify each pixel in an image, thus segmenting different regions of the image into different objects or categories, but can only capture short-range contextual information in high-resolution RS (HRRS) images. Compared to traditional CNNs, FCNs remove the last fully connected layer of CNNs to fully retain spatial information, in addition to improvements in global information retention and multiscale feature fusion.

C. Transformer-Based RSCD Methods

The transformer architecture, which has gained significant popularity in recent years, has found extensive applications in computer vision tasks, encompassing image classification [33], semantic segmentation [34], object detection [33], object tracking [35], among others. Its fundamental departure from traditional convolutional structures lies in the core concept built upon self-attention mechanisms [36], which enables the network to establish intricate relationships among elements within the input sequence, assigning weights to each element to signify its importance concerning other elements. The integration of transformers into RSCD [37], [38] has yielded notable advancements. ChangeFormer [39] unified the hierarchically structured transformer encoder with a multilayer perceptron (MLP) decoder within a single Siamese network architecture. BIT [40] consisted of one transformer encoder and two transformer decoders, enabling efficient modeling of spatio-temporal contexts. CD tasks necessitate consideration of long-range dependencies between diverse regions in an image, an area where CNNs might exhibit limitations. Transformer excels in effectively capturing long-range contextual information, showcasing its efficacy in addressing this aspect within CD frameworks. However, it falls short in capturing local information compared to RNNs and CNNs, while also incurring high computation costs.

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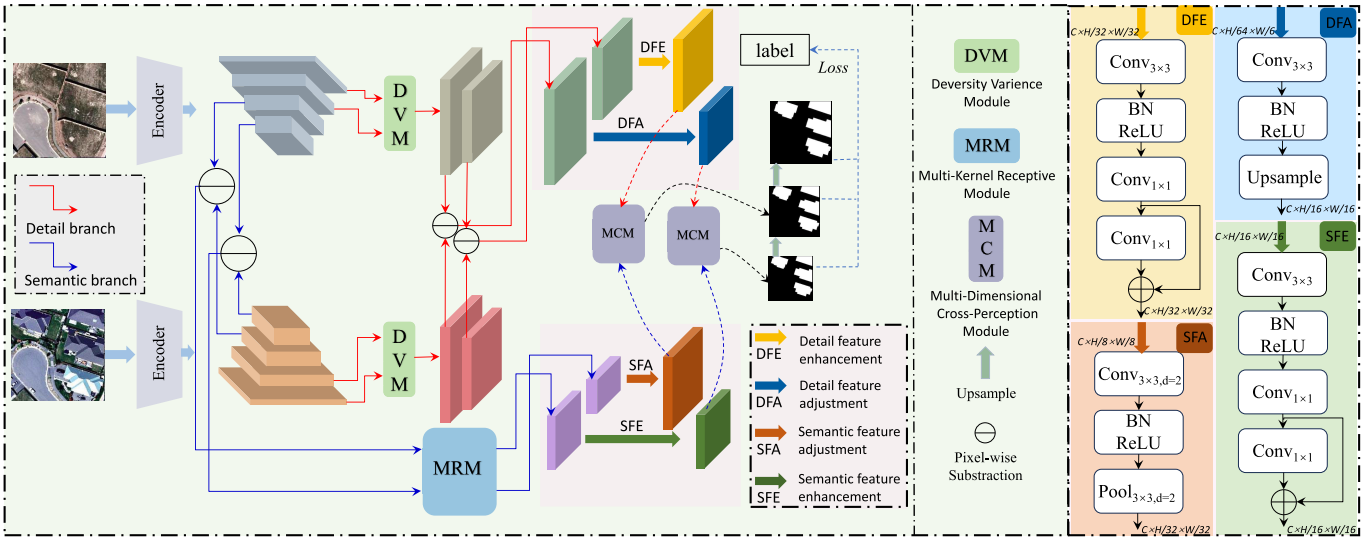


Fig. 1. Architecture of SDSN. The architecture includes detail and semantic branches that enhance feature representation. Bi-temporal images are encoded to extract multiscale features. The semantic branch processes deep-scale features, which are enriched by MRM for better context. For the detail branch, DVM focuses on noise reduction and background suppression, generating detailed change maps through feature subtraction. Both branches are enhanced, adjusted, and fused in MCM. This combination effectively captures fine-grained and contextual information, producing the final change maps via the decoder.

III. Methodology

A. Overview

The overall end-to-end deep learning architecture SDSN is proposed in this article as shown in Fig. 1. A more complete and information-rich network is provided for RSCD tasks through dual-branch feature interaction. Specifically, the lightweight CNN backbone MobileNetV2 is utilized to extract features from the original bi-temporal HRRS images to obtain hierarchical feature maps with bi-temporal $((h/4),(w/4))$, $((h/8),(w/8))$, $((h/16),(w/16))$, and $((h/32),(w/32))$ resolutions (where h stands for the height and w represents the width of input image). To address the deficiency of global information representation in CNN backbone networks, we propose the complementary feature representation achieved by the semantic branch and detail branch to accomplish a more comprehensive feature representation. The semantic branch is obtained by differentiating deep-scale features of resolutions $((h/16),(w/16))$ and $((h/32),(w/32))$ from the bi-temporal features, while the detail branch is obtained from bi-temporal features of resolutions $((h/4),(w/4))$ and $((h/8),(w/8))$. The detail branch contains parallel weight-sharing DVM modules with the same structure and parameters to remove noise and suppress background in feature pairs. Subsequently, the bi-temporal features are subtracted, resulting in detailed change maps. The semantic branch utilizes MRM to capture richer feature representations. The four output features from the semantic and detail branches are first subjected to scale adjustment and feature enhancement via detail feature enhancement (DFE), detail feature adjustment (DFA), semantic feature enhancement (SFE), and semantic feature adjustment (SFA). After these processes, the features are integrated into MCM. Leveraging guidance across different scales, MCM employs contextual information from the semantic branch to modulate the response of the detail branch and facilitate interdimensional information exchange. This fusion amalgamates fine-grained and contextual information, culminating in the generation of final change maps through the decoder.

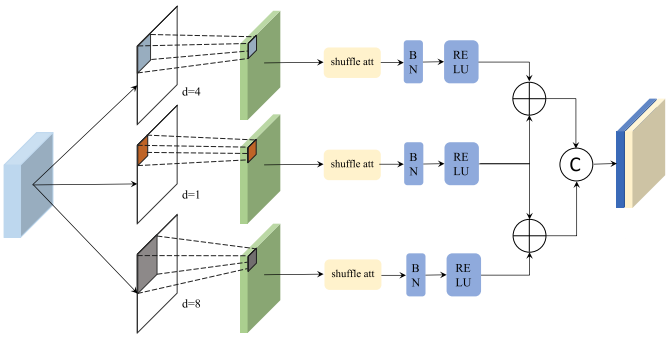


Fig. 2. Illustration of the proposed DVM. DVM consists of three branches with different atrous rates, followed by shuffle attention and activated by ReLU, and then these branches are concatenated.

B. Diversity Variance Module (DVM)

In RSCD tasks, accurately distinguishing between subtle changes in objects and background noise is a significant challenge. RS images typically feature complex backgrounds and multiscale target objects, making it difficult for traditional CNNs to effectively extract and utilize multiscale information, which often leads to inaccurate detection of changes in foreground objects and increased susceptibility to background noise. To address these challenges and obtain refined features of buildings, in detail branch, DVM is proposed as shown in Fig. 2. DVM consists of three branches, each employing atrous convolution with different atrous rates and shuffle attention [41]. These branches are concatenated in series and followed by ReLU activation, enhancing the model's ability to capture multiscale change information while effectively suppressing background noise. The input image feature map is subjected to three parallel dilated convolutions, each with dilation rates of 1, 4, and 8, respectively. This design enables the model to capture information at different scales within the same layer, effectively expanding the receptive field while

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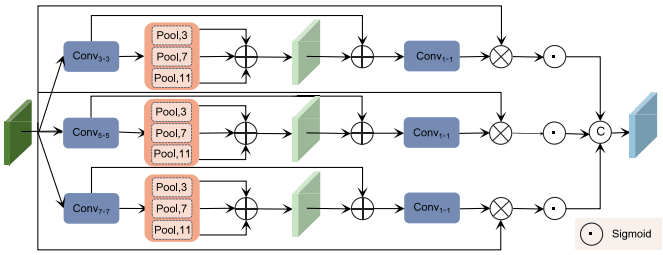


Fig. 3. Illustration of the proposed MRM. MRM utilizes parallel convolutions, multiscale pooling, and residual connections to obtain the outputs from each branch.

maintaining the spatial resolution of the feature map. The feature maps resulting from the dilated convolutions are then processed by shuffle attention, which combines channel and spatial attention mechanisms by shuffling and reorganizing the feature channels, enhancing the interaction and saliency of features. After processing each feature map with shuffle attention, batch normalization, and nonlinear transformation are applied using the ReLU activation function. ReLU truncates negative values to zero, effectively suppressing irrelevant information and noise while enhancing feature representation sparsity. This process generates the selective mapping Z_i, $i \in \{1, 2, 3\}$, $d \in \{1, 4, 8\}$

$$Z_i = \xi\left(\mathrm{SA}\left(\mathrm{Conv}_{3\times3}^{a=d}(Z)\right)\right) \tag{1}$$

where $\mathrm{Conv}_{3\times3}^{a=d}(\cdot)$ denotes a 3×3 atrous convolution with atrous rate of d, $\mathrm{SA}(\cdot)$ represents shuffle attention and ξ stands for serialization of BN and ReLU. In order to avoid excessive loss of local features, while smoothly introducing larger receptive fields to enhance overall feature representation, it ensures that local details and global information at different scales are fully processed, we conduct element-wise addition between feature maps Z_1 and Z_2, and similarly between feature maps Z_1 and Z_3. Finally, we concatenate these two summed feature maps along the channel dimension and form a comprehensive multiscale feature map Z_{out}

$$Z_{\mathrm{out}} = \mathrm{Cat}(\mathrm{Add}(Z_1, Z_2), \mathrm{Add}(Z_1, Z_3)) \tag{2}$$

where Z_{out} encapsulates multiscale information. By leveraging this progressive and mutually reinforcing strategy, the module achieves efficient feature extraction and background noise suppression.

C. Multikernel Receptive Module

To expand the receptive field and establish long-range dependencies, we propose MRM as shown in Fig. 3 to process the semantic branch. The input feature maps of this module are denoted by Q. First, the convolution operation with convolution kernel sizes of 3×3, 5×5, and 7×7 is performed, respectively, in order to capture local feature maps $\hat{Q}_i$, $i \in \{1, 2, 3\}$ at three different scales

$$\hat{Q}_i = \mathrm{Conv}_{s\times s}(Q) \tag{3}$$

where $s \in \{3, 5, 7\}$. Convolution is proficient in extracting local features but falls short in capturing broader contextual information. To fully address this, average pooling is introduced. Since single average pooling may overlook detailed information between windows due to its within-window operations, a parallel pooling module is utilized. This module performs average pooling using pooling windows of different sizes (3×3, 7×7, and 11×11) to obtain multiscale contextual information. The results from the pooling module are subsequently aggregated to consolidate feature details across various scales, thereby reinforcing the information transmission among different receptive fields and achieving an enhanced global semantic representation. Then, the results of parallel pooling are residually concatenated with $\hat{Q}_i$ to preserve the information of original features and mitigate the information loss problem. Blending is performed after 1×1 convolution and multiplying it with Q to emphasize the importance of the features learned by the module. After the sigmoid operation, the output of the single stream Q_i is obtained. Final output Q_{out} is formed by concatenating three feature streams Q_i, $i \in \{1, 2, 3\}$. MRM takes full advantage of convolution, pooling, residual connection, and feature multiplication, thereby facilitating an enriched and comprehensive feature representation

$$Q_i = \mathrm{Sig}\left(\mathrm{Conv}_{1\times1}\left(\sum_{k\in3,7,11} P_{k\times k}(\hat{Q}_i) + \hat{Q}_i\right) \times Q\right) \tag{4}$$

$$Q_{\mathrm{out}} = \mathrm{Cat}(Q_1, Q_2, Q_3) \tag{5}$$

where $i \in 1, 2, 3$. $\mathrm{Sig}(\cdot)$ represents sigmoid, a most widely used type of activation function. $\mathrm{Cat}(\cdot)$ represents feature concatenation. $P_{k\times k}(\cdot)$ represents average pooling with a kernel size of k.

D. Multidimensional Cross-Perception Module (MCM)

Through the aforementioned operations, the detail branch generates fine-grained local change features with resolutions of $((h/4), (w/4))$ and $((h/8), (w/8))$, while the semantic branch produces global change features with resolutions of $((h/16), (w/16))$ and $((h/32), (w/32))$. These feature representations are complementary, but each branch's features are independent of the other. In order to ensure that the features from the detailed branch and semantic branch are consistent and complementary in both size and semantic representation during fusion, we perform feature enhancement and scale alignment before fusing the multiscale feature maps from the dual-branches. Specifically, we designed the following modules.

1) *Detail Feature Enhancement and Semantic Feature Enhancement:* These modules use multikernel convolution stacking to enhance the features for the detail change features with a resolution of $((h/32), (w/32))$ and the semantic change features with a resolution of $((h/16), (w/16))$, respectively.
2) *Detail Feature Adjustment:* This module performs upsampling with a stride of 4 on the detail change features with a $((h/64), (w/64))$ resolution of to achieve scale alignment, resulting in an output with a resolution of $((h/16), (w/16))$.
3) *Semantic Feature Adjustment:* This module performs convolution with a stride of 2 and average pooling

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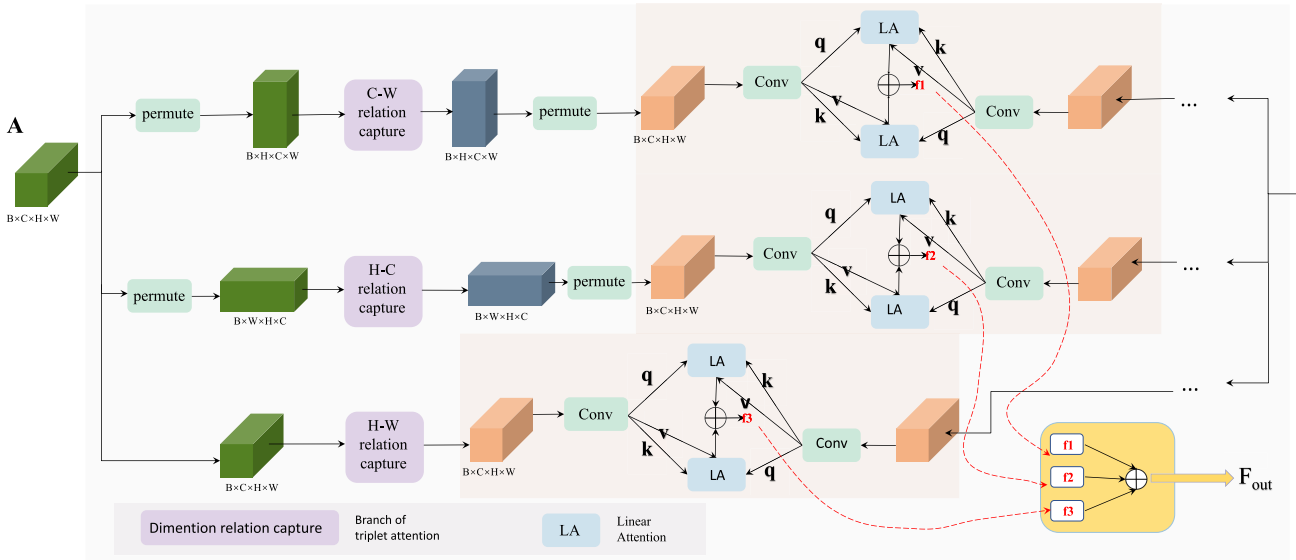


Fig. 4. Illustration of the proposed MCM. MCM captures cross-dimensional information and utilizes linear cross-attention to propagate dimensional interaction features.

with a stride of 2 on the semantic change features with a resolution of $((h/8), (w/8))$ to achieve scale alignment, resulting in an output with a resolution of $((h/32), (w/32))$.

The detail features are referred to as A, and the semantic features as B, and feed them into the network with shared weights as shown in Fig. 4.

Before the final fusion, inspired by triplet attention [42], the input features interact across dimensions to obtain three interaction features. The first branch captures the relationship between channel (C) and spatial width (W), resulting in C-W features. The second branch captures the relationship between channel (C) and spatial height (H), resulting in C-H features. The third branch captures the relationship between spatial height (H) and width (W), resulting in H-W features. In these branches, the 7×7 convolutions are replaced with 3×3 depthwise separable convolutions. This adjustment aids in reducing both the parameter count and computational cost without sacrificing information.

Given the challenges associated with traditional fusion methods in capturing intricate feature relationships, cross attention is introduced to enhance the comprehension and utilization of interaction information among these features. Specifically, for each set of dimensional interaction features derived from A and B, we organize them into teams, designated as (A_i, B_i) $(i \in \{1, 2, 3\})$, in order to facilitate deeper exploration of their mutual interactions

$$(A_i, B_i) = \psi(RC(\psi(A, B))) \tag{6}$$

where i represents 3-D interaction branches, ψ stands for rotation, and RC refers to cross-dimensional information capture. The position encoding is first performed and the channel dimensions are unified and reduced by 1×1 convolution. A_i and B_i are each obtained as two sets of vectors (Q_i^A, K_i^A, V_i^A) and (Q_i^B, K_i^B, V_i^B). One of the vectors is used as input to the query Q, while the other sequence is used as input to the key K and the value V. For each query, the degree of association between it and all the keys is computed, and the association scores are usually normalized in order to make them comparable and interpretable. This step usually consists of applying the correlation scores to the softmax function to obtain values ranging from 0 to 1. These normalized scores represent the attention weights of queries with respect to different keys, i.e., how much attention they pay to a particular key. The normalized association scores are used to weight the features in the value collection. This means that for each query, it will get the weighted information from the value collection. The weights are determined by normalized association scores from the previous step

$$\text{Cross Att} = \text{softmax}\left(\frac{Q_i^A K_i^{B^{\mathrm{T}}}}{\sqrt{d}}\right) V_i^B. \tag{7}$$

However, the operational space complexity of (7) reaches $O(N^2)$ and the time complexity reaches $O(N^2 d)$. Inspired by recent findings on attentional linearization [43], we use linear cross-attention to complete the propagation of dimensional interaction features to obtain L_i^A and L_i^B

$$L_i^A = \frac{Q_i^B}{\sqrt{d}} \text{softmax}\left(V_i^A K_i^{A^{\mathrm{T}}}\right) \tag{8}$$

$$L_i^B = \frac{Q_i^A}{\sqrt{d}} \text{softmax}\left(V_i^B K_i^{B^{\mathrm{T}}}\right) \tag{9}$$

where $i \in 1, 2, 3$, represent 3-D branches, which are summed to obtain branching output f_i

$$f_i = \text{add}\left(L_i^A, L_i^B\right). \tag{10}$$

At the end of this module, the fine-grained feature mappings produced by the three branches are aggregated by simple averaging to obtain the final output F_{out}

$$F_{\text{out}} = \text{avg}(f_1, f_2, f_3). \tag{11}$$

E. Loss Function

A prominent issue of the RSCD task is that changed regions are usually much smaller than unchanged regions, which easily

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leads to a class imbalance problem, which can adversely affect the performance and training of the model. Therefore, we use a mixture of binary cross-entropy (BCE) loss denoted as L_{bce} and dice loss denoted as L_{dice}, which can balance the performance of the model to a certain extent. L_{bce} and L_{dice} are represented by formula as

$$L_{\text{bce}}(z, q) = -\left[q \cdot \log z + (1 - q) \cdot \log(1 - z)\right] \tag{12}$$

$$L_{\text{dice}}(z, q) = 1 - \frac{2 \cdot z \cdot q}{\|z\|_1 + \|q\|_1} \tag{13}$$

where $\cdot$ represents dot-product operation, and z and q, respectively, represent the predicted change map and corresponding ground truth. $\|\cdot\|_1$ denotes the l_1 norm. The total loss L_{total} is represented as

$$L_{\text{total}}(z, q) = \sum_{i=1}^{4} (L_{\text{bce}}(z_i, q) + \lambda L_{\text{dice}}(z_i, q)) \tag{14}$$

where z_i, $i \in \{1, 2\}$ are the predicted change maps for two resolutions, respectively, and λ is a parameter that balances L_{bce} and L_{dice}.

IV. Experiments

A. Data Description

In order to assess the practical effectiveness of SDSN, we conduct comprehensive experiments on three prominent bi-temporal RSCD databases.

1) *LEVIR [12]:* The dataset comprises 637 pairs of RS images, each with dimensions of 1024 × 1024 pixels. Following the official dataset splitting setting of 7:1:2, we extracted patches of size 256 × 256, resulting in 7120 training patches, 1024 validation patches, and 2048 test patches.
2) *BCDD [11]:* It contains a pair of images of size 32 507 × 15 354. We first nonoverlapped the cropped images into 256 × 256 sized patches and then obtained 6095/762/763 image pairs for training/validation/test, respectively.
3) *SYSU [10]:* The dataset of Sun Yat-sen University consists of 20 000 pairs of dual-time-phase RS images with a spatial size of 256 × 256 and a spatial resolution of 0.5 m, randomly divided into 36 005/12 005/12 005, for training/validation/test.

B. Experimental Setup

1) Implementation Details: Our model is built based on the PyTorch framework and trained using a single NVIDIA GeForce GTX 1080 Ti with 12 GB. The image is first subjected to data enhancement on the input image block before entering the backbone network, in line with other comparative experiments. We set the initial learning rate, max iteration, and batch size as 0.0005, 100 000, and 8.

2) Evaluation Metrics: To measure the degree of similarity between predicted maps and ground truth, we employ four commonly utilized metrics for comprehensive performance evaluation: precision (Pre), recall, $F1$-score ($F1$), and intersection over union (IoU). Each metric is defined as follows:

$$\text{Pre} = \text{TP}/(\text{TP} + \text{FP}) \tag{15}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{16}$$

$$F1 = 2\text{Pre} \cdot \text{Rec}/(\text{Pre} + \text{Rec}) \tag{17}$$

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{18}$$

where TP, FP, TP, and FP denote the numbers of true positives, false positives, true negatives, and false negatives, respectively.

C. Performance Comparison

1) Comparison to SOTA: To ascertain the effectiveness and efficiency of SDSN on RSCD task, several models are selected as competitors, including CNN-based methods: FC-diff [44], FC-ef [44], FC-cat [44], SNUNet [45], ICIF [6], DMINet [47], TFI-GR [46], HANet [49], SEIFNet [48], C2FNet [50], and ELGC-Net [51], and transformer-based methods: ChangeFormer [39] and BIT [40]. We replicate these experiments using the adjusted codebase with parameters suggested by the respective studies to ensure equitable comparisons. All experiments are conducted with NVIDIA GeForce GTX 1080 Ti.

2) Quantitative Evaluation: A great quantity of experiments were completed to assess the effectiveness of the proposed model. Table I presents a quantitative evaluation of various methods across three datasets LEVIR [12], BCDD [11], and SYSU [10]. We also provide the size of the model parameters (Params) and the computational cost (Flops). The analysis results clearly demonstrate that our proposed method outperforms other competitors across all four metrics by a significant margin. Compared to TFI-GR, which exhibits relatively good detection accuracy, our approach boasts lower Params and Flops, specifically, Flops decreased by 1.86 G, and Params decreased by 24.37 M. Furthermore, when compared to other lightweight models (FC-diff, FC-ef, FC-cat, and DMINet), our method demonstrates superior detection results, achieving the highest IoU and $F1$ scores across three datasets. In summary, SDSN strikes a tradeoff between performance and cost-effectiveness.

3) Qualitative Evaluation: For a more intuitive evaluation, the different methods are compared visually on three RSCD datasets as shown in Fig. 5. Our method demonstrates superior prediction performance and excels in distinguishing between real change and pseudo change. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue). It is worth highlighting that the typical examples we have chosen each address several problems that are difficult to deal with in CD. SYSU demonstrates the shadows problem in the first row and the edge refinement problem in the second row; BCDD demonstrates the seasonal change problem in the first row and the extraneous object occlusion problem in the second row; and LEVIR demonstrates the intense light problem in the first row and the dense building detection in the second row.

D. Ablation Studies

To thoroughly assess the individual components' influence on CD performance, we undertake comprehensive ablation experiments across three datasets. The outcomes of these experiments are presented in Tables II–VI.

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TABLE I

QUANTITATIVE COMPARISON OF ARCHITECTURAL CHANGES. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND SUBOPTIMAL RESULTS IN BLUE. ALL RESULTS ARE EXPRESSED AS A PERCENTAGE (%)

Model	Flops (G)	Params (M)	Times (H)	SYSU [10]				BCDD [11]				LEVIR [12]			
				IoU	F1	Recall	Pre	IoU	F1	Recall	Pre	IoU	F1	Recall	Pre
FC-diff [44]	9.42	1.35	9.41	42.05	59.20	46.72	80.80	71.94	83.68	77.76	90.57	75.18	85.83	84.59	87.12
FC-ef [44]	7.13	1.35	9.44	55.83	71.65	65.41	79.22	60.13	75.10	67.85	84.09	71.80	83.59	82.27	84.94
FC-cat [44]	10.62	1.55	9.52	60.68	75.53	71.38	80.21	71.19	83.19	78.50	88.49	76.98	86.99	85.64	88.38
SNUNet [45]	246.22	27.07	22.62	66.89	79.87	79.39	80.18	81.62	89.61	86.29	93.47	82.59	89.95	88.58	91.39
BIT [40]	16.88	3.01	13.71	62.50	76.93	72.50	81.93	81.16	89.60	89.97	89.24	82.06	90.15	88.98	91.34
ChangeFormer [39]	10.63	3.50	54.37	60.46	75.36	71.06	80.22	77.15	87.10	85.27	89.01	81.93	90.07	87.38	92.92
ICIF [6]	25.36	23.82	41.79	61.38	76.07	73.31	79.04	81.07	89.51	86.67	92.55	79.86	88.80	86.85	90.84
TFI-GR [46]	9.73	28.37	5.33	71.38	83.30	81.16	85.56	88.89	94.12	94.10	94.14	83.03	90.73	89.53	91.95
DMINet [47]	6.42	14.42	15.41	68.20	81.10	76.82	85.88	80.90	89.42	91.88	87.13	83.14	90.79	88.89	92.78
SEIFNet [48]	8.37	27.91	10.26	69.91	82.28	79.85	85.12	87.88	92.31	92.57	90.04	83.37	90.66	88.97	92.87
HANet [49]	14.07	3.03	24.93	62.35	76.81	72.98	80.01	86.02	91.35	93.37	94.03	81.86	89.74	89.32	91.14
C2FNet [50]	62.10	16.1	13.4	64.82	78.77	75.70	81.10	88.62	94.18	92.75	95.23	84.69	91.71	89.11	94.46
ELGC-Net [51]	123.59	10.57	9.39	65.34	79.03	77.26	80.89	86.95	93.11	93.62	93.71	83.52	91.02	89.78	92.30
Ours	7.94	4.03	7.25	72.08	83.89	83.81	83.72	89.83	94.82	92.78	96.33	84.41	91.58	89.97	93.37

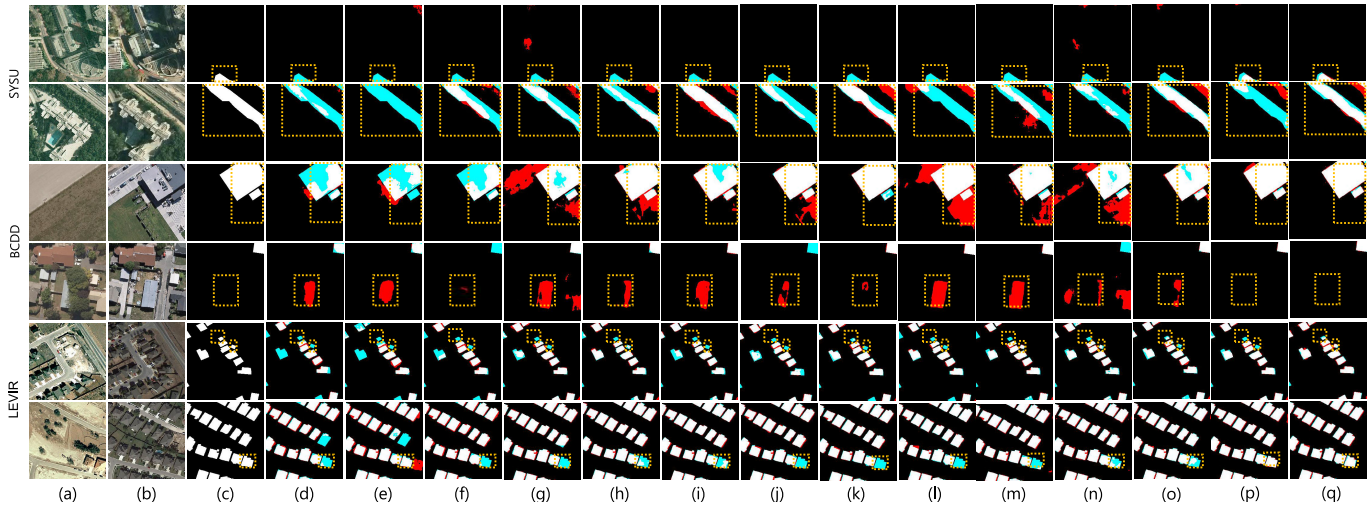


Fig. 5. Visual comparisons on SYSU, BCDD, and LEVIR dataset. (a) t1 image. (b) t2 image. (c) Ground truth. (d) FC-Diff. (e) FC-ef. (f) FC-cat. (g) SNUNet. (h) BIT. (i) ChangeFormer. (j) ICIF. (k) TFI-GR. (l) DMINet. (m) ELGC-Net. (n) SEIFNet. (o) HANet. (p) C2FNet. (q) Ours. The areas with significant improvements over the compared methods are highlighted and displayed using yellow dashed boxes.

1) Effectiveness of DVM: DVM module aims at suppressing noise in the extracted obtained feature. To validate the effectiveness of DVM, we substitute it with a 3 × 3 convolutional layer having identical output channels as DVM and call this method (a) wo-DVM in Table II, the best results have been highlighted in bold. The removal of DVM resulted in a significant performance degradation compared to the original model, proving the superiority of DVM. Furthermore, to demonstrate that the best performance can be obtained with our chosen atrous rates, in Fig. 6, different atrous rates are set, such as (b) {1, 2, 4} and (c) {1, 3, 6}. It can be observed that the original setting {1, 4, 8} can perform the RSCD task better. In addition, to demonstrate the effectiveness of the proposed method for merging three dilation rate branches, we compared it with two alternative methods: (d) direct summation of three branches, and (e) direct concatenation of three branches. Experimental results validated the importance of smooth transitions. To further demonstrate the functionality of the DVM module, we visualize the heatmaps as shown in Fig. 6, (a) and (b) correspond to three pairs of bi-temporal pictures, (c) and (e) show the feature pictures extracted from backbone, which contains a large amount of noise, making it difficult to distinguish between the foreground and background

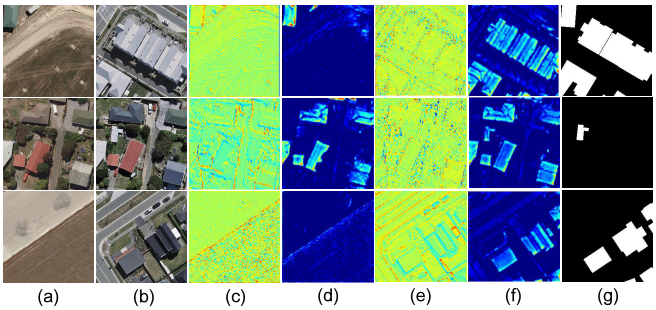


Fig. 6. Visualization of heat maps for DVM module. (a) t1 image. (b) t2 image. (c) Input of DVM module for t1 image. (d) Output of DVM module for t1 image. (e) Input of DVM module for t2 temporal image. (f) Output of DVM module for t2 image. (g) Ground truth.

as they blend together, (d) and (f) are output maps of DVM module, where noise is effectively suppressed, emphasizing valuable information and clearly separating the foreground from the background.

2) Effectiveness of MRM: We propose an MRM module to capture feature information at different scales to improve multilevel contextual understanding and information synthesis. To demonstrate its effectiveness, we remove the MRM module and substitute it with a 3 × 3 convolutional layer with an

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TABLE II
ABLATION STUDIES OF DVM WHEN TRAINED ON THREE CD DATASETS. $F1$ IS REPORTED IN PERCENTAGE (%)

No.	Method	SYSU	BCDD	LEVIR
(a)	w/o DVM	82.58	93.84	90.53
(b)	{1,2,4}	82.62	93.33	91.31
(c)	{1,3,6}	83.31	94.25	89.88
(d)	sum	83.14	93.52	90.21
(e)	concat	82.79	93.97	89.69
(f)	Ours	**83.89**	**94.82**	**91.58**

TABLE III
ABLATION STUDIES OF MRM WHEN TRAINED ON THREE CD DATASETS. $F1$ IS REPORTED IN PERCENTAGE (%)

No.	Method	SYSU	BCDD	LEVIR
(a)	w/o MRM	82.81	93.39	90.39
(b)	{1,3,5}	83.58	93.98	91.39
(c)	{3,6,9}	82.43	94.55	91.43
(d)	Ours	**83.89**	**94.82**	**91.58**

identical output channel as MRM, and call this method wo-MRM [(a) in Table III], the best results have been highlighted in bold. Based on experimental outcomes, it is inferred that substituting the MRM module results in a decline in the detection accuracy of $F1$ metric for SYSU by approximately 1.81%. In addition, we set different convolutional kernel sizes to complete the average pooling module, e.g., (b) {1, 3, 5} and (c) {3, 6, 9}, with the original kernel size {3, 5, 7} to obtain the best performance.

To further demonstrate the functionality of MRM, we can intuitively observe the visualization results in Fig. 7. (a) shows the prechange image, (b) represents the postchange image, (c) displays the change map before MRM, (d) illustrates the change map after MRM, and (e) shows the ground truth. The yellow dashed boxes effectively highlight the beneficial role of MRM in capturing more comprehensive change features. In the first row, roads are mistakenly identified as buildings. In the second row, the boundaries of the shed roof are difficult to distinguish from the ground. In the third row, the houses at the image boundary are embedded into the farmland, making it challenging for the model to recognize them. MRM can effectively combine global semantics to accurately distinguish between buildings and nonbuildings.

3) Effectiveness of Dual-Branch: In this experiment, the semantic branch is removed directly and called this method as wo-SS [(b) in Table IV], the best results have been highlighted in bold. The detection accuracy of $F1$ for SYSU is reduced by about 1.27%, which proves the effectiveness of semantics in supplementing the detailed information.

To further demonstrate the practicality of the dual-branch, we conduct visualization experiments shown in Fig. 8. (a) shows prechange images, (b) represents postchange images, (c) displays detail change maps, (d) highlights semantic change maps, (e) represents dual-branch fused change maps, and (f) shows ground truth. The yellow dashed boxes highlight areas of interest.

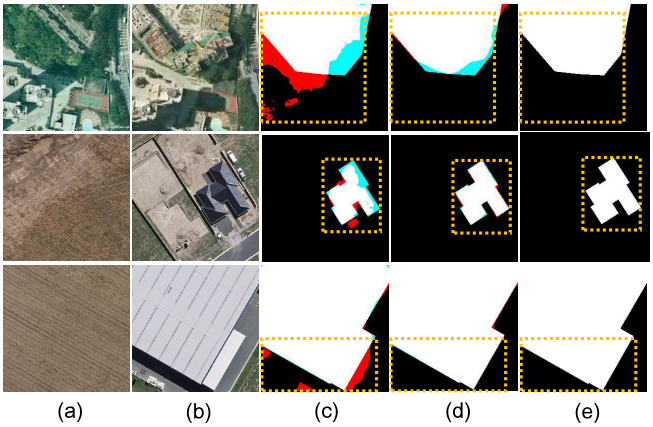


Fig. 7. Visualization of ablation experiments for MRM. (a) t1 image. (b) t2 image. (c) Before MRM. (d) After MRM. (e) Ground truth. The areas with significant improvements are highlighted and displayed using yellow dashed boxes.

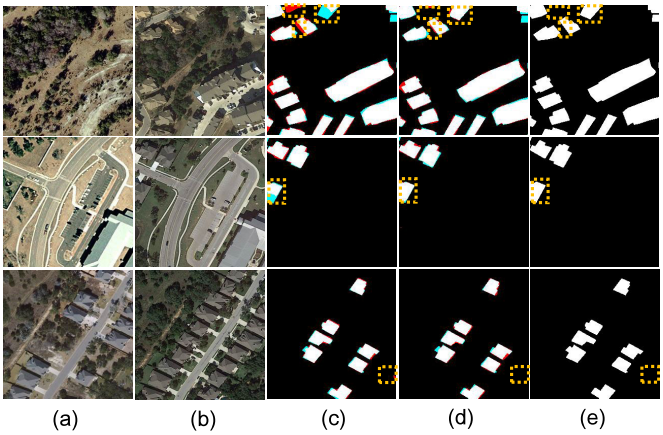


Fig. 8. Visualization of ablation experiments for dual-branch. (a) t1 image. (b) t2 image (c) Detail change maps. (d) Semantic change maps. (e) dual-branch fused change maps. (f) Ground truth. The areas with significant improvements are highlighted and displayed using yellow dashed boxes.

In the first row, the detail branch struggles to distinguish houses merged with leaves, while the semantic branch effectively differentiates them. The fused map separates the houses from irrelevant factors. In the second row, the detail branch correctly identifies the haystack as a nonchange area, whereas the semantic branch misidentifies it as a building. The fused map suppresses nonbuilding changes. In the third row, the detail branch identifies the left boundary house corner as nonchange but fails on the right boundary. The semantic branch fails on the left but succeeds on the right. The fused map combines both branches' strengths for more accurate identification. These fully demonstrate the significance of the dual-branches capturing complementary features from different perspectives.

4) Effectiveness of MCM: We introduce MCM for feature fusion of the detail branch with the semantic branch. To demonstrate the superiority of this module, we replace it with a simple summation to complete the fusion task. And call this method as wo-MCM [(a) in Table IV], the best results have been highlighted in bold. From experimental results, it is concluded that when the MCM is replaced, the detection accuracy of $F1$ for SYSU is reduced by about 1.52%, and it

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TABLE IV
ABLATION STUDIES OF MCM AND DUAL-BRANCH WHEN TRAINED ON THREE CD DATASETS. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

No.	Method	SYSU	BCDD	LEVIR
(a)	w/o MCM	82.37	93.96	90.41
(b)	w/o SS	82.62	93.82	90.88
(c)	Ours	**83.89**	**94.82**	**91.58**

TABLE V
ABLATION STUDIES OF DIFFERENT BACKBONES WHEN TRAINED ON THREE CD DATASETS. $F1$ IS REPORTED IN PERCENTAGE (%)

No.	Method	SYSU	BCDD	LEVIR	Params(M)	Flops(G)
(a)	ResNet 18	83.26	93.26	90.74	23.9	36.29
(b)	ResNet 50	**83.92**	94.64	91.52	129.46	101.13
(c)	Ours	83.89	**94.82**	**91.58**	**4.03**	**7.94**

TABLE VI
ABLATION STUDIES OF λ WHEN TRAINED ON THREE CD DATASETS. $F1$ IS REPORTED IN PERCENTAGE (%)

No.	Method	SYSU	BCDD	LEVIR
(a)	λ= 0	82.81	93.39	90.39
(b)	λ= 0.25	83.58	93.98	91.39
(c)	λ= 0.5	82.43	94.55	91.43
(d)	λ= 1	83.81	94.76	91.51
(e)	Ours	**83.89**	**94.82**	**91.58**

is difficult to fuse the exhaustive feature information with the general fusion method.

5) Discussions of Different Backbones: In order to verify the efficiency of MobileNetV2 backbone, we use different backbones ResNet18 and ResNet50 for comparison (a) and (b) in Table V, the best results have been highlighted in bold. The detection accuracy of $F1$ for the SYSU dataset is approximately similar. Params/Flops but increased by 19.87 M/28.35 G and 125.43 M/93.19 G, respectively.

6) Discussions of λ: In order to balance BCE loss and dice loss, we conducted experiments with different values of λ to determine the optimal choice. As shown in Table VI, the best results have been highlighted in bold, when λ is set to 0.75, the detection accuracy is the highest, indicating that the balance of loss is optimal at this value.

V. CONCLUSION

The proposed SDSN presents a comprehensive approach for RSCD. By leveraging the complementary strengths of the semantic and detail branches, SDSN effectively captures both fine-grained local changes and global contextual changes. First, our method uses MobileNetV2 to extract multilevel features. After denoising the shallow bi-temporal features using DVM, detail change maps are derived through differential operation. Similarly, semantic change maps are obtained by differentiating deep bi-temporal features. The MRM module is then utilized to capture multiscale global information effectively. Subsequently, through operations like feature enhancement and scale alignment, MCM facilitates cross-dimensional information exchange, integrating detail change maps and semantic change maps. MCM provides a comprehensive representation of change information from multiple perspectives. The experimental results demonstrate the effectiveness of SDSN across different datasets, showcasing its robustness and superior performance compared to existing methods, along with fewer parameters and lower computational costs.

However, existing advanced CD methods typically rely on a large amount of finely detailed pixel-level annotated data. C2F-SemiCD [50] proposes a semi-supervised learning method that uses consistency regularization to enhance the feature learning capability of both labeled and unlabeled data, it demonstrates significant advantages in terms of training and inference time efficiency. Ensuring detection accuracy while achieving semi-supervised and unsupervised RSCD tasks is the goal of our next research step.

REFERENCES

[1] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008, doi: 10.1080/01431160801950162.

[2] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018, doi: 10.1109/TGRS.2018.2858817.

[3] J. Tian, S. Cui, and P. Reinartz, "Building change detection based on satellite stereo imagery and digital surface models," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 406–417, Jan. 2014, doi: 10.1109/TGRS.2013.2240692.

[4] F. Bovolo and L. Bruzzone, "A split-based approach to unsupervised change detection in large-size multitemporal images: Application to tsunami-damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1658–1670, Jun. 2007, doi: 10.1109/TGRS.2007.895835.

[5] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636, doi: 10.1016/j.rse.2021.112636.

[6] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213, doi: 10.1109/TGRS.2022.3168331.

[7] C.-P. Chen, J.-W. Hsieh, P.-Y. Chen, Y.-K. Hsieh, and B.-S. Wang, "SARAS-net: Scale and relation aware Siamese network for change detection," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 12, pp. 14187–14195, doi: 10.1609/aaai.v37i12.26660.

[8] Z.-H. You et al., "Crossed Siamese vision graph neural network for remote-sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4409916, doi: 10.1109/TGRS.2023.3325536.

[9] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[10] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5604816, doi: 10.1109/TGRS.2021.3085870.

[11] R. E. Kennedy et al., "Remote sensing change detection tools for natural resource managers: Understanding concepts and tradeoffs in the design of landscape monitoring projects," *Remote Sens. Environ.*, vol. 113, no. 7, pp. 1382–1396, Jul. 2009, doi: 10.1016/j.rse.2008.07.018.

[12] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020, doi: 10.3390/rs12101662.

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[13] A. Singh, "Change detection in the tropical forest environment of northeastern India using Landsat," *Remote Sens. Tropical Land Manage.*, vol. 44, pp. 254–273, Mar. 1986, doi: 10.22215/etd/2004-06089.

[14] W. J. Todd, "Urban and regional land use change detected by using Landsat data," *J. Res. U.S. Geol. Surv.*, vol. 5, no. 5, pp. 529–534, 1977.

[15] X. Dai and S. Khorram, "Quantification of the impact of misregistration on the accuracy of remotely sensed change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. Remote Sens.–Sci. Vis. Sustain. Develop.*, vol. 4, Aug. 1997, pp. 1763–1765, doi: 10.1109/IGARSS.1997.609062.

[16] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 2, pp. 125–133, Nov. 2006, doi: 10.1016/j.isprsjprs.2006.09.004.

[17] D. K. Seo, Y. H. Kim, Y. D. Eo, M. H. Lee, and W. Y. Park, "Fusion of SAR and multispectral images using random forest regression for change detection," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 10, p. 401, Oct. 2018, doi: 10.3390/ijgi7100401.

[18] G. Verdier and A. Ferreira, "Adaptive Mahalanobis distance and k-nearest neighbor rule for fault detection in semiconductor manufacturing," *IEEE Trans. Semicond. Manuf.*, vol. 24, no. 1, pp. 59–68, Feb. 2011, doi: 10.1109/TSM.2010.2065531.

[19] G. F. Byrne, P. F. Crapper, and K. K. Mayo, "Monitoring land-cover change by principal component analysis of multitemporal Landsat data," *Remote Sens. Environ.*, vol. 10, no. 3, pp. 175–184, Nov. 1980, doi: 10.1016/0034-4257(80)90021-8.

[20] S. Jin and S. A. Sader, "Comparison of time series tasseled cap wetness and the normalized difference moisture index in detecting forest disturbances," *Remote Sens. Environ.*, vol. 94, no. 3, pp. 364–372, Feb. 2005, doi: 10.1016/j.rse.2004.10.012.

[21] D. Xue, T. Lei, X. Jia, X. Wang, T. Chen, and A. K. Nandi, "Unsupervised change detection using multiscale and multiresolution Gaussian-mixture-model guided by saliency enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1796–1809, 2021, doi: 10.1109/JSTARS.2020.3046838.

[22] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2223–2232, doi: 10.1109/ICCV.2017.244.

[23] Z. Jin et al., "Mining contextual information beyond image for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7231–7241, doi: 10.1109/ICCV48922.2021.00714.

[24] A. M. El Amin, Q. Liu, and Y. Wang, "Zoom out CNNs features for optical remote sensing change detection," in *Proc. 2nd Int. Conf. Image, Vis. Comput. (ICIVC)*, Jun. 2017, pp. 812–817, doi: 10.1109/ICIVC.2017.7984667.

[25] Y. Cui, L. Yan, Z. Cao, and D. Liu, "TF-blender: Temporal feature blender for video object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8138–8147, doi: 10.1109/ICCV48922.2021.00803.

[26] I. Sutskever, J. Martens, and G. E. Hinton, "Generating text with recurrent neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2011, pp. 1017–1024.

[27] A. Graves, A.-R. Mohamed, and G. E. Hinton, "Speech recognition with deep recurrent neural networks," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, Mar. 2013, pp. 6645–6649, doi: 10.1109/ICASSP.2013.6638947.

[28] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[29] Y. Wang, "Polyphonic sound event detection with weak labeling," Ph.D. thesis, 2018.

[30] X. Tang et al., "An unsupervised remote sensing change detection method based on multiscale graph convolutional network and metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021, doi: 10.1109/TGRS.2021.3106381.

[31] J. Wang, F. Gao, J. Dong, S. Zhang, and Q. Du, "Change detection from synthetic aperture radar images via graph-based knowledge supplement network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1823–1836, 2022, doi: 10.1109/JSTARS.2022.3146167.

[32] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666, doi: 10.1109/CVPR.2019.00075.

[33] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022, doi: 10.1109/ICCV48922.2021.00986.

[34] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890, doi: 10.1109/CVPR46437.2021.00681.

[35] X. Chen, B. Yan, J. Zhu, D. Wang, X. Yang, and H. Lu, "Transformer tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8126–8135.

[36] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514715, doi: 10.1109/TGRS.2021.3115699.

[37] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021, doi: 10.1109/TGRS.2021.3130716.

[38] L. Bashmal, Y. Bazi, and M. A. Rahhal, "Deep vision transformers for remote sensing scene classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2815–2818, doi: 10.1109/IGARSS47720.2021.9553684.

[39] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210, doi: 10.1109/IGARSS46834.2022.9883686.

[40] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[41] Q.-L. Zhang and Y.-B. Yang, "SA-Net: Shuffle attention for deep convolutional neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 2235–2239, doi: 10.1109/ICASSP39728.2021.9414568.

[42] M. Guo et al., "Attention mechanisms in computer vision: A survey," *Comput. Vis. Media*, vol. 8, no. 3, pp. 331–368, 2022, doi: 10.1007/s41095-022-0271-y.

[43] W. Xu, Y. Xu, T. Chang, and Z. Tu, "Co-scale conv-attentional image transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9981–9990, doi: 10.1109/ICCV48922.2021.00983.

[44] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067, doi: 10.1109/CVPR.2018.00474.

[45] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021, doi: 10.1109/LGRS.2021.3056416.

[46] Z. Li, C. Tang, L. Wang, and A. Y. Zomaya, "Remote sensing change detection via temporal feature interaction and guided refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628711, doi: 10.1109/TGRS.2022.3199502.

[47] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015, doi: 10.1109/TGRS.2023.3241257.

[48] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414, doi: 10.1109/TGRS.2024.3360516.

[49] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3867–3878, 2023, doi: 10.1109/JSTARS.2023.3264802.

[50] C. Han, C. Wu, M. Hu, J. Li, and H. Chen, "C2F-SemiCD: A coarse-to-fine semi-supervised change detection method based on consistency regularization in high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4702621, doi: 10.1109/TGRS.2024.3370568.

[51] M. Noman, M. Fiaz, H. Cholakkal, S. Khan, and F. S. Khan, "ELGC-Net: Efficient local–global context aggregation for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701611, doi: 10.1109/TGRS.2024.3362914.

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